AI.307

Machine Vision

COURSE OVERVIEW

Advanced machine vision: 3D reconstruction, object tracking, and vision systems for industrial and robotic inspection.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: VISION FOR MACHINES

KEY TOPICS

- Machine vision vs. computer vision
- Industrial vision systems
- Vision pipeline
- Applications

TEACHING FOCUS

Engineer vision for inspection and guidance.

KEY TAKEAWAY Machine vision measures and decides.

MACHINE VISION

2

SESSION

60 MINUTES

3D VISION

KEY TOPICS

- Structured light and stereo
- Time-of-flight
- 3D reconstruction
- Point clouds

TEACHING FOCUS

Recover 3D from 2D images.

KEY TAKEAWAY Depth makes vision spatial.

3

SESSION

60 MINUTES

OBJECT TRACKING

KEY TOPICS

- Tracking by detection
- Kalman filter tracking
- Optical flow tracking
- Occlusion handling

TEACHING FOCUS

Follow objects over time.

KEY TAKEAWAY Tracking links frames into stories.

4

SESSION

60 MINUTES

INSPECTION & GAUGING

KEY TOPICS

- Defect detection
- Measurement and gauging
- Barcode and OCR
- Vision-guided robotics

TEACHING FOCUS

Automate quality inspection.

KEY TAKEAWAY Vision inspects faster and more reliably than people.

MACHINE VISION

5

SESSION

60 MINUTES

VISION SYSTEMS DESIGN

KEY TOPICS

- Lighting and optics
- Camera selection
- Real-time constraints
- Integration

TEACHING FOCUS

Build a complete vision system.

KEY TAKEAWAY Lighting and optics determine image quality.

6

SESSION

60 MINUTES

MACHINE VISION & REVIEW

KEY TOPICS

- Vision system demo
- Course recap and assessment
- Toward intelligent control (AI.308)
- Inspection task

TEACHING FOCUS

Deploy vision on the factory floor.

KEY TAKEAWAY Good vision makes good products.